

An Overview of Force Measurement

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Abstract—The abstract goes here.

Index Terms—Interaction Force, Force Measurement.

I. INTRODUCTION

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II. FORCE SENSING METHODS

A. Resistive force transducer

B. Capacitive force transducer

C. Inductive force transducer

III. STATIC VS DYNAMIC FORCE MEASUREMENT

Mechanical force measurements are classified into static and dynamic measurements based on the time variation or frequency spectrum of the applied load or force. Force measurements are classified as static when the force remains constant over time or changes slowly. Examples include weight measurement[1] and static strain testing in structures. Conversely, force measurements are dynamic when the force value changes rapidly over time such as those encountered in robotics and dynamic calibration requiring transducers with high frequency response capabilities. When discussing dynamic and static force measurements it is essential to note that the physical measurement technique varies depending on the frequency of the force being measured. It is common practice to use resistive strain gauges for measuring static forces and piezoelectric strain gauges for dynamic forces [2]. In static force measurements piezoelectric transducers exhibit slight drift because of inherent charge leakage across the insulation resistance—governed by the charge amplifiers RC discharge time constant inevitably leads to low frequency drift whereas resistive strain gauges show no significant drift. Resistive strain gauges have high stiffness which ensures minimal mechanical creep and supreme baseline stability under static loads. Although piezoelectric transducers can be used to measure large static forces, they are preferred over resistive strain gauges for dynamic force measurements. Piezoelectric materials like quartz have low material density and high dynamic frequency response which make them the superior choice for dynamic regimes. Piezoelectric quartz elements possess a significantly lower mass density $\rho = 2650\text{kg/m}^3$

compared to copper $\rho = 8900\text{kg/m}^3$. Since the natural frequency of a structural system is inversely proportional to the square root of its mass ($f \propto \sqrt{K/M}$) the drastic mass reduction in quartz effectively shifts its resonance spectrum upward. To substantiate the preceding findings we conducted a Finite Element Analysis (FEA) using Solidworks software. A 100N force was applied to two geometrically identical beams ($10 \times 100 \times 10$) mm one made of copper and the other of quartz. Subjecting both to a static load as illustrated in Fig. 1. The copper beam characterized by isotropic properties exhibited a maximum displacement of $6.22\text{ }\mu\text{m}$ whereas the anisotropic quartz beam showed a greater deformation of $8.211\text{ }\mu\text{m}$. Depending on the displacement values in both cases we calculated the stiffness of the copper and quartz beams to be 16.06 MN/m and 12.318 MN/m , respectively based on the relationship (1).

$$K = \frac{F}{d} \quad (1)$$

Harmonic frequency response analysis was conducted to evaluate the dynamic behavior of both materials within a frequency range of approximately 100 kHz as shown in Fig .(2). Copper exhibited a distinct resonance peak at the fundamental frequency 3432.2 HZ , while its response remained flat—and virtually negligible—across the rest of the frequency range, this behavior is attributed to its high damping coefficient. In contrast, quartz displayed secondary resonance peaks at harmonic frequencies throughout the studied range, alongside a primary resonance peak at its fundamental frequency 5652.7 HZ , owing to a small damping coefficient that is one hundred times lower than that of copper. The simulation results confirm that quartz is the best choice for the dynamic measurement of mechanical forces and moments. The fig .3 demonstrates the relative advantages of using resistive versus piezoelectric strain gauges, based on the measured force values and the duration of measurement [2]. Some research indicates the feasibility of using piezoelectric transducers to measure static forces as in [3],[4]. The concept of static and dynamic force measurement can be extended to the concept of static and dynamic torque measurement. The Fig .4 shows dynamic and static torque. Torque measurement is considered dynamic when the transducer or sensing element is mounted on the rotating shaft or component whose torque is being measured. A dynamic torque transducer typically provides two primary outputs: rotational torque and rotational speed. Electrical signals from dynamic transducers are transmitted to the data acquisition system in one of two ways: via slip rings and brushes for low-speed rotation or wirelessly for high-speed rotation. Examples of dynamic torque measurement include monitoring the torque and output power of high-speed gas or steam turbines. The measured torque is considered static (reaction torque) when the angular acceleration of the rotating element is negligible

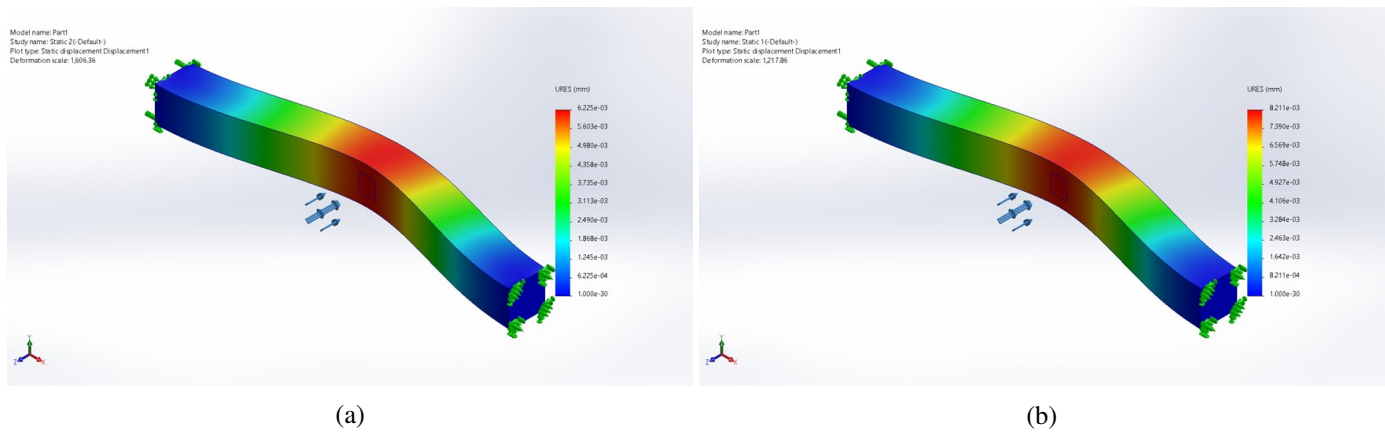


Fig. 1: Static Analysis in FEA (a): Displacement of copper beam. (b): Displacement of Quartz beam.

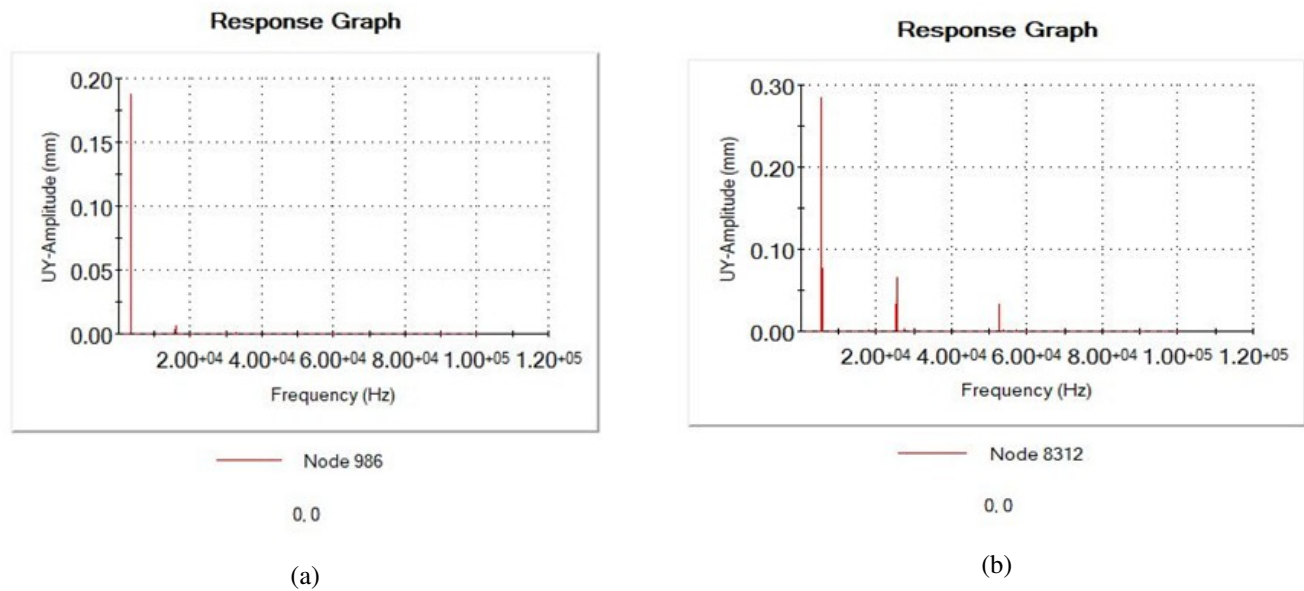


Fig. 2: Frequency response analysis in FEA (a): Copper beam. (b): Quartz beam.

or zero. The electrical signal resulting from this static torque measurement is transmitted via a simple connection (cable). A primary application of static torque measurement is quality control testing—such as measuring the torque required to rotate a stationary steering column [5].

IV. MULTI-DIMENSIONAL FORCE MEASUREMENT

V. FORCE ESTIMATION

VI. FORCE MEASUREMENT APPLICATIONS

A. Force measurement in monitoring and testing tasks

B. Use of force measurement in automated systems

C. Force measurement in research and development laboratories

VII. CONCLUSION

The conclusion goes here.

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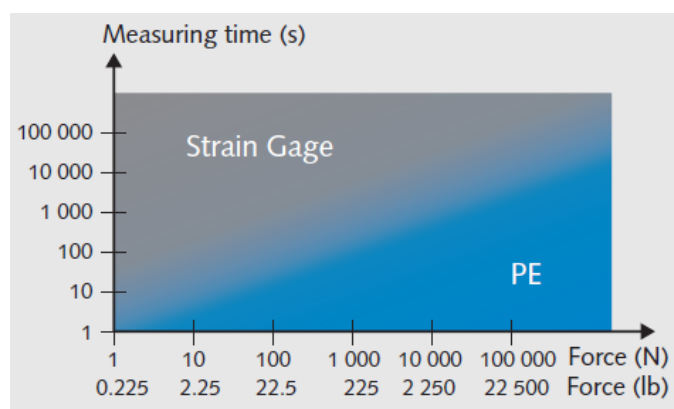


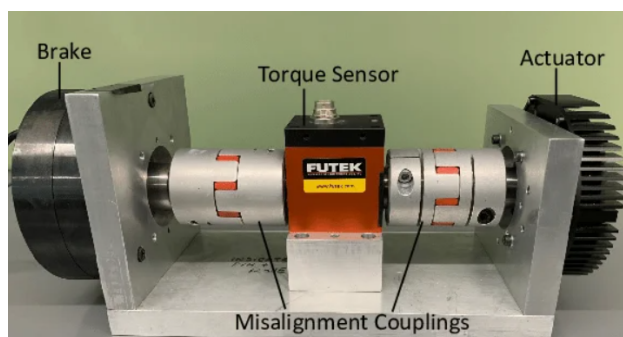
Fig. 3: Measuring times and force ranges: piezoelectric (PE) versus strain gauge (basis: drift ± 0.05 pC/s and measurement error of 1%)



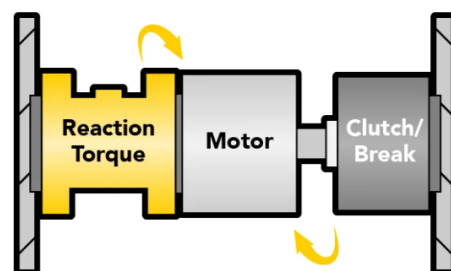
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(a)



(b)

Fig. 4: (a):Dynamic torque sensor; (b): Static torque sensor.